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### METHOD FOR PRODUCING PSICOSE BY FERMENTATION AND ISOMERIZATION OF BACILLUS SUBTILIS

#### Abstract

The present invention relates to the technical field of fermentation, and provides a method for producing psicose by fermentation and isomerization of *Bacillus subtilis*. According to the method of the present invention, fermentation parameters are strictly controlled, and catalytic reaction is performed by using recombinant *Bacillus subtilis* obtained by fermentation as enzyme liquid, so that the conversion rate of psicose in a system is improved, the problems of low enzyme catalytic reaction efficiency and low yield are solved, and the method is simple to operate and environment-friendly.

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## Background/Summary

### REFERENCE TO SEQUENCE LISTING

[0001] The instant application contains a Sequence Listing in XML format as a file named "GWPCT22278-seql.xml", created on Nov. 21, 2022, of 16,834 bytes in size, and which is hereby incorporated by reference in its entirety.

### TECHNICAL FIELD

[0002] The present invention relates to the technical field of fermentation, and particularly relates to a method for producing psicose by fermentation and isomerization of *Bacillus subtilis*.

### BACKGROUND

[0003] D-psicose (D-allulose or D-psicose, hereinafter referred to as psicose) is a rare sugar which is present in small amounts in wheat, murraya, and processed sugar cane and sugar beet syrups; in addition, psicose is also present in a small amount in a mixed syrup of fructose and glucose. In the heat treatment of high-sugar foods, psicose may be produced by conversion from fructose by a non-enzymatic reaction, and thus it is also present in a trace amount in processed foods such as sauces, juices, sugar cane and beet syrup.

[0004] Psicose has many specific physiological properties, compared with glucose, psicose has a lower absorption rate in human body, and competes with glucose and fructose at cell membrane surface transporters, thereby reducing fructose and glucose absorption in the daily diet. The special physiological properties of psicose make it of great significance to human health, such as: enhancing insulin resistance, inhibiting postprandial blood glucose increase, reducing fat accumulation in abdomen, and preventing diabetes. In addition, psicose has antiinflammatory, nerve protecting, active oxygen free radical scavenging, and arteriosclerosis treating effects. In 2014, psicose was recognized by the U.S. food and drug administration as a safe food (GRAS). The special physiological properties of psicose make it important for use in the fields of medicine and health food.

[0005] At present, the production method of psicose comprises a biological enzyme method and a fermentation method, but the fermentation method has low enzyme activity, so that the yield is low, the operation is complex, the production cost is high, and the requirement of industrial production cannot be met.

### SUMMARY

[0006] The invention aims to provide a method for producing psicose by fermentation and isomerization of *Bacillus subtilis*, which can greatly improve the activity of D-psicose-3-epimerase produced by the *Bacillus subtilis* so as to improve the conversion rate of the psicose, simplifies the operation and is suitable for popularization and application in industrial large-scale production of psicose.

[0007] The invention provides a method for producing psicose by fermentation and isomerization of *Bacillus subtilis*, which comprises the following steps: [0008] Inoculating recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing D-psicose 3-epimerase into a fermentation culture medium for fermentation culture to obtain fermentation liquid; during the fermentation culture period, the aeration ratio is 0.1-2 V/V.Math.min; the rotation speed of the fermentation culture is 200-1000 rpm; [0009] Separating thalli from the fermentation liquid; [0010] Carrying out catalytic reaction on a substrate fructose by the thalli to obtain the psicose.

[0011] Preferably, the fermentation medium comprises the following components: 10-40 g/L of bean cake powder, 20-50 g/L of bran, 30-60 g/L of corn flour, 1-10 g/L of dipotassium hydrogen

phosphate, 1-10 g/L of magnesium sulfate and 0.1-5 g/L of manganese sulfate, wherein the pH value of the fermentation medium is 6-8.

[0012] Preferably, the fermentation medium comprises the following components: 20-35 g/L of bean cake powder, 35-45 g/L of bran, 45-55 g/L of corn flour, 4-8 g/L of dipotassium hydrogen phosphate, 5-8 g/L of magnesium sulfate and 1-3 g/L of manganese sulfate, wherein the pH value of the fermentation medium is 6.5-7.5.

[0013] Preferably, the rotation speed of the fermentation culture is 200-1000 rpm, and the aeration ratio is 0.1-2 V/V.Math.min; [0014] The temperature of the fermentation culture is 30-37° C., and

the time of the fermentation culture is 48-72 h; [0015] During fermentation, the pH value of fermentation system is controlled to be 6-8; [0016] The inoculation amount of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is 1%-10%.

[0017] Preferably, the system of the catalytic reaction is 1-5 ml of suspension of the thalli, 50-150 g of fructose and 50-200 ml of water; [0018] The thalli concentration of the thalli suspension is (30-48)×10<sup>9</sup> CFU/ml. [0019] Preferably, the temperature of the catalytic reaction is 50-60° C.; [0020] The time of the catalytic reaction is 2-4 h.

[0021] Preferably, the culture medium of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is an LB culture medium; [0022] The culture temperature of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-

epimerase is 30-37° C.; [0023] The culture time of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is 20-24 h; [0024] The rotation speed of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is

200-400 rpm; [0025] When the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is cultured, the aeration ratio is 0.1-1 V/V.Math.min; [0026] The inoculation amount of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is 1%-10%.

[0027] Preferably, the method for separating the thalli is ceramic membrane ultrafiltration; [0028] The pore diameter of the ceramic membrane for ultrafiltration is 0.22 nm or less.

[0029] The invention provides a method for producing psicose by fermentation and isomerization of *Bacillus subtilis*, wherein recombinant *Bacillus subtilis* 1A751-DPE obtained by fermentation is used as an enzyme solution to perform catalytic reaction on fructose, and the aeration ratio and the rotation speed during fermentation culture are strictly controlled, so that the enzyme activity of D-psicose 3-epimerase of the recombinant *Bacillus subtilis* 1A751-DPE is favorably improved, and the conversion rate of psicose is improved. The results of the examples show that the highest conversion rate of the psicose in the system can reach 31.4%; the method effectively solves the problems of low efficiency and low yield of the enzyme catalysis reaction, is simple to operate and environment-friendly, and has important market value and application value. Meanwhile, the host bacteria in the invention are *Bacillus subtilis*, so that the food safety performance is high.

[0030] The method provided by the invention specifically limits the fermentation conditions and the culture medium, and has the following advantages:

[0031] According to the method for preparing the psicose, the fermentation is carried out by one-time feeding, so that the catalytic reaction efficiency is improved, the operation steps are reduced, the production cost is reduced, the industrial production is easy, and the problems of low enzyme activity, complex operation and high production cost of the existing fermentation method are solved; in addition, the bean cake powder, the bran, the corn flour and the like used in the fermentation medium are cheap and widely available raw materials, so that the production cost is greatly reduced. The fermentation method provided by the invention catalyzes the production of psicose, the enzyme activity level high, the conversion rate is over 29.3% and is far higher than the existing reported yield, the production cost is reduced, and the industrial production is facilitated.

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## Description

### BRIEF DESCRIPTION OF FIGURES

[0032] FIG. 1 is a schematic diagram showing chromosomal *dal* gene knock-out of *Bacillus subtilis*;

[0033] FIG. 2 shows the result of verifying the loss of the thermo-sensitive plasmid pTSC;

[0034] FIG. 3 shows the results of detection of psicose in example 2;

[0035] FIG. 4 shows the results of detection of psicose in example 3;

[0036] FIG. 5 is the result of detection of psicose in example 4.

### DETAILED DESCRIPTION

[0037] The invention provides a method for producing psicose by fermentation and isomerization of *Bacillus subtilis*, which comprises the following steps:

[0038] Inoculating recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing D-psicose 3-epimerase into a fermentation culture medium for fermentation culture to obtain fermentation liquid; during the fermentation culture period, the aeration ratio is 0.1-2 V/V.Math.min; the rotation speed of the fermentation culture is 200-1000 rpm; [0039] Separating thalli from the fermentation liquid; [0040] Carrying out catalytic reaction on a substrate fructose by the thalli to obtain the psicose.

[0041] According to the invention, recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing D-psicose 3-epimerase is inoculated into a fermentation medium for culture to obtain fermentation liquid.

[0042] In the present invention, recombinant *Bacillus subtilis* 1A751-DPE for expressing D-psicose 3-epimerase is constructed by preferably knocking out the *dal* gene (expressing D-alanine racemase) of *Bacillus subtilis* 1A751 by homologous recombination technique, and simultaneously transferring the recombinant plasmid pUB-P43dpe-dal expressing D-psicose 3-epimerase (DPE), and the recombinant *Bacillus subtilis* 1A751-DPE was developed to efficiently express D-psicose 3-epimerase.

[0043] In the present invention, the culture medium for the recombinant *Bacillus subtilis* 1A751-DPE seed liquor expressing the D-psicose 3-epimerase is preferably LB medium. The culture temperature of the recombinant *Bacillus subtilis* 1A751-DPE seed liquor expressing the D-psicose 3-epimerase is preferably 30-37° C., more preferably 32-35° C., and most preferably 33° C. The culture time of the recombinant *Bacillus subtilis* 1A751-DPE seed liquor expressing the D-psicose 3-epimerase is preferably 20-24 h, more preferably 21-23 h, and most preferably 22 h. The culture rotation speed of the recombinant *Bacillus subtilis* 1A751-DPE seed liquor expressing the D-psicose 3-epimerase is preferably 200-400 rpm, more preferably 250-350 rpm, and most preferably 300 rpm. When the recombinant *Bacillus subtilis* 1A751-DPE seed liquor expressing the D-psicose 3-epimerase is cultured, the aeration ratio is preferably 0.1-1 V/V.Math.min, more preferably 0.2-0.8 V/V.Math.min, most preferably 0.4-0.7 V/V.Math.min, and most preferably 0.5 V/V.Math.min. The aeration ratio is the ratio of the volume of air passing through a unit volume of culture solution per minute. The inoculation amount of the recombinant *Bacillus subtilis* 1A751-DPE seed liquor expressing the D-psicose 3-epimerase is preferably 1%-10%, more preferably 2%-8%, further preferably 4%-7%, and most preferably 5%. The preparation of the recombinant *Bacillus subtilis* 1A751-DPE seed liquor expressing the D-psicose 3-epimerase is preferably carried out in a seed tank. The volume of the seed tank is preferably 30-70 L, and more preferably 50 L.

[0044] In the present invention, the fermentation medium preferably comprises the following components: 10-40 g/L of bean cake powder, 20-50 g/L of bran, 30-60 g/L of corn flour, 1-10 g/L of dipotassium hydrogen phosphate, 1-10 g/L of magnesium sulfate and 0.1-5 g/L of manganese sulfate, wherein the pH value of the fermentation medium is 6-8, and the fermentation medium preferably comprises the following components: 20-35 g/L of bean cake powder, 35-45 g/L of bran,

45-55 g/L of corn flour, 4-8 g/L of dipotassium hydrogen phosphate, 5-8 g/L of magnesium sulfate and 1-3 g/L of manganese sulfate, wherein the pH value of the fermentation medium is 6.5-7.5. [0045] In the present invention, the rotation speed of the fermentation culture is preferably 200-1000 rpm, more preferably 400-800 rpm, and further preferably 500 rpm. The aeration ratio is preferably 0.1-2 V/V.Math.min, more preferably 0.2-1.8 V/V.Math.min, still more preferably 0.5-1.5 V/V.Math.min, and most preferably 1.0 V/V.Math.min. The temperature of the fermentation culture is preferably 30-37° C., more preferably 32-35° C., and most preferably 34° C. The time for fermentation culture is preferably 48-72 h, more preferably 52-68 h, further preferably 54-65 h, and most preferably 60 h. During the fermentation, the pH value of the fermentation system is preferably controlled to be 6-8, more preferably 6.5-7.5, and most preferably 7.0. The inoculation amount of the recombinant *Bacillus subtilis* 1A751-DPE seed liquor expressing the D-psicose 3-epimerase is preferably 1%-10%, more preferably 2%-8%, even more preferably 4%-7%, and most preferably 5%. The cultivation is preferably carried out in a fermenter. The volume of the fermenter is preferably 250-750 L, more preferably 350-700 L, and further preferably 500 L.

[0046] After the fermentation liquid is obtained, the invention separates the thalli from the fermentation liquid.

[0047] In the present invention, the method for separating the thalli is preferably ceramic membrane ultrafiltration. The pore diameter of the ceramic membrane for the ceramic membrane ultrafiltration is preferably 0.22  $\mu$ m or less, more preferably 0.2  $\mu$ m.

[0048] After obtaining the thalli, the thalli are subjected to catalytic reaction on a substrate fructose to obtain the psicose.

[0049] In the present invention, the system of the catalytic reaction is preferably 1-5 ml of suspension of the thalli, 50-150 g of fructose and 50-200 ml of water; the water is preferably deionized water. The thalli concentration of the thalli suspension is  $(30-48) \times 10^9$  CFU/ml, and more preferably  $42 \times 10^9$  CFU/ml. The temperature of the catalytic reaction is preferably 50-60° C., more preferably 52-58° C., and most preferably 55° C. The time of the catalytic reaction is preferably 2-4 h, and more preferably 3 h.

[0050] In the present invention, the fermentation liquid is centrifugally filtered to remove impurities, transferred into a liquid phase small bottle and analyzed by high performance liquid chromatography, and the analysis conditions of the high performance liquid chromatography are as follows: chromatographic column: Carbpmix Ca-NP; mobile phase: ultrapure water (pre-sonication to exclude air); flow rate: 0.6 mL/min; column temperature: 85° C.; detector: differential refractometer: temperature of the detector: 30° C.; sample injection amount: 10  $\mu$ L. Analyzing the proportion of the psicose peak area according to a peak diagram area normalization method of liquid chromatogram to obtain the conversion rate of the psicose.

[0051] The method for producing psicose by fermentation using *Bacillus subtilis* provided by the present invention is described in detail below with reference to examples, but they should not be construed as limiting the scope of the present invention.

#### EXAMPLE 1

##### Construction Method of Recombinant *Bacillus subtilis* 1A751-DPE

[0052] 1. The primers required in the construction process are shown in Table 1, and the primers are synthesized by Sangon Biotech (Shanghai) Co., Ltd.

TABLE-US-00001 TABLE 1 Primers required for the construction procedure

| Primers | Sequence (5'.fwdarw.3')                                          |
|---------|------------------------------------------------------------------|
| P1      | TACAAAGCAAAAGCGAAAATGACCATC (SEQ ID NO: 1)                       |
| P2      | ATGCTATACGAACGGTACTTAGAAGAGAGAAGGTAA (SEQ ID NO: 2)              |
| P3      | TTACCTTCCTTTTAAGTACCGTTCGTATAGCAT (SEQ ID NO: 3)                 |
| P4      | CAAGCAAAGCTGTTTTATCTACCGTTCGTATAATGT (SEQ ID NO: 4)              |
| P5      | ACATTATACGAACGGTAGATAAAACAGCTTTGCTTG (SEQ ID NO: 5)              |
| P6      | CAGCTGATAGGATTCTTGCTCGCTTA (SEQ ID NO: 6)                        |
| P7      | CGTCTGTACGTTCTTAAGGAATTCTTAGGAGTGTTTATGACATTCTAAT (SEQ ID NO: 7) |

NO: 7) P8 TAGAATGCAAAAAGTGAAATCATAATGATAGGTGGTATGTTTTTCGCTTGA  
(SEQ ID NO: 8) P9  
ATTAGAATGTCATAAACACTCCTAAGAATTCCTTAAGGAACGTACAGACG (SEQ ID  
NO: 9) P10 TCAAGCGAAAACATAACCACCTATCATTATGATTTCACCTTTTGCATT  
(SEQ ID NO: 10) P11  
ATTAACCTACCTAAATGGAGAATTCAATCTATTATTAATCTGTTTCAGCAA (SEQ ID  
NO: 11) P12 CACCATTTCCTTTAGAAAGATCTTCCCTTTTCAGATAATTTTAGATTT  
(SEQ ID NO: 12) P13  
AAATCTAAAATTATCTGAAAAGGGAAGATCTTTCTAAAGAGGAAATGGTG (SEQ ID  
NO: 13) P14 TTGCTGAACAGATTAATAATAGATTGAATTCTCCATTAGGTAAGTTAAT  
(SEQ ID NO: 14)

## 2. Construction of Recombinant *Bacillus subtilis* 1A751(dal-)

### 2.1 Extraction of *Bacillus subtilis* 1A751 Genome

[0053] The extraction of the genome of *B. subtilis* 1A751 refers to an operation manual of an Ezup column type bacterial genome DNA extraction kit (Sangon Biotech (Shanghai) Co., Ltd.), and the specific steps are as follows:

[0054] (1) A single *B.subtilis* 1A751 colony was picked from a fresh LB plate, inoculated into 4 mL of LB liquid medium, and cultured overnight on a shaker at 37° C. at 200 r/min.

[0055] (2) 1 mL of *B.subtilis* 1A751 bacterial liquid was sucked, and the bacterial was  
[0056] separated by centrifugation at 8000 r/min for 1 min. Added 180 µL of 20 mg/mL lysozyme solution prepared in advance to resuspend the thalli, carried out water bath at 37° C. for 60 min, then added 400 µL of Buffer Digestion, shaken and mixed uniformly. Finally, the mixture was kept in a water bath at 65° C. for 60 min (evenly mixed by reversing every 10 min in the process) until the mixture became clear and transparent (cells were completely lysed).

[0057] (3) Added 20 µL of 10 mg/mL RNaseA, and stood at room temperature for 2-5 min.

[0058] (4) Added 200 µL Buffer PB and mixed well by inversion, -left at -20° C. for 5 min.

[0059] (5) Centrifugation was carried out at 10000 r/min for 5 min, and the supernatant was transferred to a new 1.5 mL centrifuge tube.

[0060] (6) Added isopropanol with the same volume, and reversed for 5-8 times to fully and uniformly mix. Standed at room temperature for 2-3 min, centrifuged at 10000 r/min for 5 min, and discarded the supernatant.

[0061] (7) Suspended and precipitated with 1ml of 75% ethanol, rinsed by inversion for 1-3 min, centrifuged at 10000 r/min for 2 min, and discarded the supernatant.

[0062] (8) Repeated step (7) once.

[0063] (9) Opened the cover and stood for 5-10 min to completely volatilize the residual ethanol.

[0064] (10) Dissolved with 50 µL TE Buffer (10 mmol/L Tris/HCl, 1 mmol/L EDTA, pH 8.0). The resulting DNA solution can be subjected to the next experiment or stored at -20° C.

### 2.2 Amplification of Homology Arms

[0065] Fragments with the length of 800-900 bp are respectively selected from two sides of a D-alanine racemase gene (GenBank No. CAB12271.1) on a chromosome of the *Bacillus subtilis* 1A751 to be used as homologous arms, and amplification primers are designed. The upstream homology arm (SEQ ID NO:15,

tacaaagcaaaagcgaaaatgaccatcgagacagggagtgagcctcaagtatacaatgtggaaatctggcacaaaaagccgtct  
ctgtacaggggtttattagaaaatccgaaaaaggaccaaaccagtcattttacgaaatgaaaacggagattcgttctcacgccg  
tctttaataagagcttcagatttcaaagcgattggccgaacaacagcagccaggtttatctgtttgaatcgctcgtaaaggatgttc  
aaaacgattcggatgcagttttcacagcgaaagaaaagaatacgtatttgaacaaaaaactatcagcataacaaaatgct  
gcccacgcaagaaatcacatttaacaaaaaagatatgagcccgctcatctgtcaaagtgtgatactgaccggaaagtgtggtca  
aagtagaattcagcagctttgaatttaataaacaatttgataaagaatcatttgatgaaaagaaaaatgaccctttctcaaatgg  
atgtcgccacaagcgcaaaagccttcgatacatttgcggtcaaaacgcccgtggaactgccgcttgccgtcaagctgctgaagaa  
aaagatatctactgaagacgggaagcgcgcacatcatgacgtacggaggagaaaaatcatttacgtaattcaggaaaaagccc

agattgccaaaggctctcctcctgctgaacggcggaaccgggtaaatctcggtacaccatcggcgccctgtcggatgcatcatt  
atcatggacatatgacggcgtagattaccttctcttcttaa) was amplified by the primer pair P1 and P2, and the  
downstream homology arm (SEQ ID NO:16,  
ataaaacagctttgcttgaagagtgaataatggtatcattatcactggatggataaaagagtcgtatccgtaagtttggtggagggtgt  
atgtttttgtctgaatccagcgcaagaaccgaaatgaaaatcagcttgcggaaaacctagtagctgaattggatgggttagcgat  
gcgggagaaacgaagcagaaacgaactgatcacaaagcagtgagagcgtatgtcagcgaacgaacaactcgtcacaaccgtga  
tttgatgagacgcggctatatggaaatggcgaaaatcaacctgaattttcttctgaggctcactttgcggagtgcgaggctgaaac  
gacagttgagcgcttagtcagcggaggataatcatttgattgtgaaacgcggcgatgtttattttgctgattatctcctgtgttggt  
cagagcaaggcggggtgcgcccgttttagtgatccaaaatgacatcggaatcgcttcagcccaactgctattgttgagccataa  
cagcacaatacagaaagcgaaattaccaaccacgctcgaaatcgatgaaaacgctacgggtttgaaagagattccgtattttg  
ctggagcaaattcggacgattgacaagcaaaggtaacggataagattactcatctggatgatgaaatgatggataaggtgatga  
agccttacaaatcagtttggcactcattgatttttagacatatttgagggtgctcaaatagagcaactttttgtttcaaaaaacat  
aaacgatataatagtgaataacgaaaaaatatgtgtttttattgggaggtgaagcgaatttgatgtcgaaccagactgtatacca  
gttcattgccgaaaatcaaatgaactgctccaactgtggactgacacactaaaagaattaagcgagcaagaatcctatcagctg) was  
amplified by the primer pair P5 and P6, using the genome of *Bacillus subtilis* 1A751 as a template.  
PCR amplification products are respectively recovered and purified and the DNA concentration  
was measured.

#### Amplification of the lox71-spc-lox66 Resistance Gene Expression Cassette

[0066] An upstream primer P3 and a downstream primer P4 are respectively designed according to  
the sequence of the lox71-spc-lox66 resistance gene expression cassette, and the primers P3 and P4  
are respectively reverse complementary with the sequences of the primers P2 and P5. The plasmid  
p7S6 is used as a template, primers P3 and P4 are used for amplification to obtain a lox71-spc-  
lox66 resistance gene expression cassette, and PCR amplification product gel is recovered and  
purified, and the DNA concentration is measured.

#### 1.4 Fusion of lox71-spc-lox66 Resistance Gene Expression Cassette and Upstream, Downstream Homology Arms

[0067] The fusion of the three DNA fragments adopts a two-step PCR method (Shevchuk N A et al,  
2004), and the specific operation steps are as follows:

[0068] The first PCR reaction system is shown in Table 2.

TABLE-US-00002 TABLE 2 PCR reaction System Phusion HF Buffer (5×) 5 μL dNTP Mixture  
(2.5 mmol/L each) 2 μL Upstream homologous arm fragments (40 ng/μL) 2.5 μL lox71-spc-lox66  
fragment (40 ng/μL) 0.5 μL Downstream homologous arm fragments 2.5 μL (40 ng/μL) Phusion  
High-Fidelity DNA Polymerase 0.5 μL ddH.sub.2O Supplement the system with 25 μL

#### PCR Reaction Conditions

[0069] Performing pre-denaturation at 98° C. for 2 min; melting at 98° C. for 10 s, annealing at 56°  
C. for 15 s, and extending at 72° C. for 2 min for 10 cycles; extending at 72° C. for 5 min.

[0070] The second PCR reaction system is shown in Table 3.

TABLE-US-00003 TABLE 3 PCR reaction System Phusion HF Buffer (5×) 10 μL dNTP Mixture  
(2.5 mmol/L each) 4 μL Primer P1 1.25 μL Primer P6 1.25 μL PCR product in the first step 2 μL  
Phusion High-Fidelity DNA Polymerase 0.5 μL ddH.sub.2O Supplement the system with 50 μL

#### PCR Reaction Conditions

[0071] Performing pre-denaturation at 98° C. for 2 min; melting at 98° C. for 10 s, annealing at 60°  
C. for 15 s, and extending at 72° C. for 2 min for 30 cycles; extending at 72° C. for 5 min.

#### 2.5 Preparation and Transformation of Competence of *Bacillus subtilis*

[0072] *Bacillus subtilis* competent cells were prepared by a two-step (GM I, GM II) method  
(Julkowska D et al, 2005). The specific operation steps are as follows:

[0073] (1) The *B.subtilis* strain stored in an ultralow temperature refrigerator was streaked on a  
spore slant, and placed in an incubator at 37° C. for static culture for 24-48 h.

[0074] (2) A loopful of bacterial liquid was picked and inoculated into 5 ml of GM I medium and  
cultured overnight in a shaker at 37° C. at 200 r/min.

[0075] (3) The next day, 2 mL of GM I culture was transferred to 18 mL of GM I medium and cultured at 37° C. at 200 r/min for 3.5 h.

[0076] (4) 10 mL of the culture was transferred to 90 mL of GM II medium, cultured at 37° C. for 1.5 h at 100 r/min, and centrifuged at 6000 r/min for 10 min to separate the thalli.

[0077] (5) The thalli suspension was gently suspended in 10 ml of supernatant, and the mixed solution was dispensed into sterile 1.5 mL centrifuge tubes at 500 µL/tube and used directly for transformation, or 50% glycerol was added to a final concentration of 10% and stored immediately in a -80° C. ultra-low temperature freezer.

[0078] (6) During transformation, putting the centrifugal tube into a water bath at 45° C. for melting, a proper amount of recombinant plasmid or DNA fragment was added, and gently mixed.

[0079] (7) Resuscitated at 37° C. at 100 r/min for 30 min, spreaded plate containing corresponding antibiotics, and cultured in 37° C. incubator by inverting overnight.

## 2.6 Screening of Recombinant *Bacillus subtilis* 1A751(dal-, SPCR)

[0080] *Bacillus subtilis* 1A751 competence was prepared according to the method of step 2.5, the fusion fragment obtained in step 2.4 is transformed into the competence, and finally spreaded on plates to which both D-alanine and spectinomycin were added, and transformants were picked the next day and spotted on LB plates and plates to which both D-alanine and spectinomycin were added for further selection.

## 2.7 Knock-Out of Antibiotic Resistance Gene

[0081] (1) Using recombinant *Bacillus subtilis* 1A751(dal-, SPCR) as a host bacterium, competence was prepared in reference to step 2.5, and D-alanine solution was additionally added to the competent medium GM I and GM II to give a final concentration of 200 µg/mL.

[0082] (2) The plasmid pTSC was transformed into the above-mentioned *Bacillus subtilis* competence, spreaded on LB plates containing erythromycin and D-alanine, and subjected to inverted culture at 37° C. for 12 h in an incubator.

[0083] (3) Transformants were selected from the plate, and inoculated into an LB liquid culture medium containing erythromycin and D-alanine, performed shake culture at 37° C. by using a shaking table, and added IPTG for induction for 6 h when OD<sub>sub.600</sub> is 0.6-0.8. A loopful of bacterial liquid was picked up using an inoculating ring, and marked on an LB plate which is free of the antibiotics and added with D-alanine.

[0084] (4) Single colonies were selected from the overnight-cultured plate, and dropped on LB plate added with D-alanine and LB plate simultaneously added with spectinomycin and D-alanine, and carried out overnight culture to screen out colonies which do not grow on the spectinomycin plate and grow on the corresponding LB plate containing D-alanine.

[0085] (5) The obtained recombinant bacterium was cultured for 2 days at 51° C., so that the temperature-sensitive plasmid pTSC was lost. The resulting D-alanine auxotrophic mutant strain was designated *B.subtilis* 1A751 (dal-).

## 3. Introduction of Foreign Plasmid

### 3.1 Construction of Recombinant Plasmid

[0086] The plasmid pUB110 contains the Hpa II promoter, a *Bacillus subtilis* endogenous promoter P43 was added based on the Hpa II promoter. Recombinant plasmid construction procedures refer to the "Simple cloning" method (You et al, 2012). The specific operation steps are as follows:

[0087] (1) Amplified the target gene of P43-dpe. The plasmid T-P43-dpe was used as a template, primers P7 and P8 were used for amplification, and PCR product gel was recovered and purified and its DNA concentration was measured.

[0088] (2) The vector skeleton was amplified with primers P9 and P10 using plasmid pUB110 as template, PCR product gel was recovered and purified and its DNA concentration was measured.

[0089] (3) The reagents in table 4 were added to a microcentrifuge tube, and PCR reaction conditions are as follows:

[0090] Performing pre-denaturation at 98° C. for 2 min; melting at 98° C. for 10 s, annealing at 60°



C. for 15 s, and extending at 72° C. for 2 min for 30 cycles; extending at 72° C. for 5 min.  
TABLE-US-00004 TABLE 4 PCR reaction System Phusion HF Buffer (5×) 10 μL dNTP Mixture (2.5 mmol/L each 5 μL P43-dpe gel recycled fragments 2 μL Vector skeleton gel recycled fragment 2 μL (100 ng/μL) Phusion High-Fidelity DNA Polymerase 0.5 μL H.sub.2O Supplement the system with 50 μL

[0091] (4) The PCR product (multimeric fragment) was transformed into *Bacillus subtilis* 1A751 competence, and plated on LB plate containing kanamycin for selection.

[0092] (5) Transformants were picked and inoculated into LB liquid medium, and plasmids were extracted the next day. Plasmids were extracted from *Bacillus subtilis*, referring to a small extraction kit (Sangon Biotech (Shanghai) Co., Ltd.) operation manual of plasmid DNA, suspending thalli by using Buffer P1 in the first step, adding 20 mg/ml lysozyme, keeping the temperature at 37° C. for 30 min, then adding Buffer P2, and the subsequent steps are the same as the operation manual. The recombinant plasmid obtained by extraction was designated pUB-P43dpe.

### 3.2 Replacement of Antibiotic Resistance Gene by D-Alanine Racemase Gene

[0093] (1) Recombinant plasmid pUB-P43dpe was used as a template, primers P11 and P12 were used to amplify the vector skeleton sequence of the antibiotic resistance gene removed part, and the obtained PCR product gel was recovered and purified.

[0094] (2) Primers P13 and P14 were used to amplify the D-alanine racemase gene by using the *B.subtilis* 1A751 genome as a template, and PCR product gel was recovered and purified.

[0095] (3) The D-alanine racemase gene and the corresponding vector skeleton fragment are subjected to PCR, and a recombinant plasmid based on the dal gene screening marker is constructed by using “Simple cloning” method. Fragments of the multimer were transformed into D-alanine deficient *Bacillus subtilis* 1A751(dal-) competence, transformed into 1A751(dal-) strain competence due to the fact that the plasmid carried the dal gene, and recombinant plasmid selection was achieved by selection on LB plates without the addition of D-alanine.

[0096] (4) The resulting recombinant plasmid was designated pUB-P43dpe-dal.

### 3.3 Construction of Recombinant *Bacillus subtilis*

[0097] In the experiment, recombinant plasmids were directly constructed in *Bacillus subtilis*, so that the positive clone transformant containing the recombinant plasmids, which is obtained by screening in step 3.2, is the recombinant *Bacillus subtilis* 1A751/PUB-P43dpe-dal, and the transformant is designated 1A751-DPE.

### EXAMPLE 2

[0098] A method for producing psicose by fermentation and isomerization of *Bacillus subtilis* comprises the following specific implementation steps:

[0099] (1) Seed preparation: inoculating the *Bacillus subtilis* implanted with the *Closridium* gene prepared in example 1 into an LB medium, and culturing for 20 h at 37° C. and 100 rpm;

[0100] (2) Cultivation in a 50 L seed tank: inoculating the *Bacillus subtilis* obtained in step (1) into a seed tank filled with 35 L of LB culture medium according to the inoculation amount of 5% of the volume ratio for culturing, controlling the rotation speed of the seed tank to be 300 rpm, controlling the aeration ratio to be 0.5 V/V.Math.min, and culturing at 35° C. for 20 h to obtain seed liquid;

[0101] (3) Production in a fermentation tank: inoculating the seed liquid into a fermentation tank filled with a fermentation culture medium for culture according to the inoculation amount of 5% volume ratio, wherein the preparation method of the fermentation culture medium comprises the steps of mixing 10 g of bean cake powder, 20 g of bran, 30 g of corn flour, 1 g of dipotassium phosphate, 1 g of magnesium sulfate and 0.1 g of manganese sulfate, adjusting the pH to 7 by using a sodium hydroxide solution, and fixing the volume to 1000 ml by using deionized water; controlling the rotation speed of the fermentation tank to be 500 rpm, controlling the aeration ratio to be 1 V/V.Math.min, controlling the pH to be 7.5 at the temperature of 30° C., culturing for 60 h,

and obtaining fermentation liquid after fermentation;

[0102] (4) Carrying out ultrafiltration on the fermentation liquid obtained in step (3) by adopting a ceramic membrane with a pore diameter of 0.22  $\mu\text{m}$  to obtain *Bacillus subtilis* thalli and obtain enzyme liquid;

[0103] (5) An isomeric system: 5 ml of the enzyme solution was mixed with 114 g of fructose and 109 ml of deionized water, followed by reaction at 55° C. for 4 h. And after the reaction is finished, heating the reaction solution to inactivate enzyme, centrifugally filtering the sample, performing liquid chromatography analysis treatment, and analyzing the area ratio of the psicose peak according to a peak map area normalization method of the liquid chromatography to obtain the conversion rate of the psicose.

[0104] The technical scheme is adopted to produce psicose by fermentation, and the conversion rate of psicose is 29.3%.

#### EXAMPLE 3

[0105] A method for producing psicose by fermentation and isomerization of *Bacillus subtilis* comprises the following specific implementation steps:

[0106] (1) Seed preparation: inoculating the *Bacillus subtilis* implanted with the *Closridium* gene prepared in example 1 into an LB medium, and culturing for 22 h at the temperature of 35° C. and rotation speed of 200 rpm;

[0107] (2) Cultivation in a 50 L seed tank: inoculating the *Bacillus subtilis* obtained in step (1) into a seed tank filled with 35 L of LB culture medium according to the inoculation amount of 8% of the volume ratio for culture, controlling the rotation speed of the seed tank to be 400 rpm, controlling the aeration ratio to be 1 V/V.Math.min, and culturing at the temperature of 30° C. for 24 h to obtain seed liquid;

[0108] (3) Production in a 5000 L fermentation tank: inoculating seed liquid into a fermentation tank filled with 3500 L of fermentation medium for culture according to the inoculation amount of 8% volume ratio, wherein the preparation method of the fermentation medium comprises the steps of mixing 20 g of bean cake powder, 40 g of bran, 55 g of corn flour, 4 g of dipotassium hydrogen phosphate, 6 g of magnesium sulfate and 3.5 g of manganese sulfate, adjusting the pH to 7.2 by using a sodium hydroxide solution, and fixing the volume to 1000 ml by using deionized water; controlling the rotation speed of the fermentation tank to be 500 rpm, controlling the aeration ratio to be 1.5 V/V.Math.min, controlling the pH to be 8 at the temperature of 30° C., culturing for 72 h, and obtaining fermentation liquid after the fermentation is finished;

[0109] (4) Carrying out ultrafiltration on the fermentation liquid obtained in step (3) by adopting a ceramic membrane with a pore diameter of 0.22  $\mu\text{m}$  to obtain *Bacillus subtilis* thalli and obtain enzyme liquid;

[0110] (5) An isomeric system: 5 ml of the enzyme solution was mixed with 114 g of fructose and 109 ml of deionized water, followed by reaction at 60° C. for 3 h, and the psicose conversion was measured and calculated according to the method of example 2. The result shows that the conversion rate of the psicose produced by the technical scheme is 29.8%.

#### EXAMPLE 4

[0111] A method for producing psicose by fermentation and isomerization of *Bacillus subtilis* comprises the following specific implementation steps:

[0112] (1) Seed preparation: inoculating the *Bacillus subtilis* implanted with the *Closridium* gene prepared in example 1 into an LB medium, and culturing for 24 h at the temperature of the LB medium temperature of 37° C. and rotation speed of 250 rpm;

[0113] (2) Cultivation in a 50 L seed tank: inoculating the *Bacillus subtilis* obtained in step (1) into a seed tank filled with 35 L of LB culture medium according to the inoculation amount of 8% of the volume ratio for culturing, controlling the rotation speed of the seed tank to be 400 rpm, controlling the aeration ratio to be 0.3 V/V.Math.min, and culturing at the temperature of 30° C. for 24 h to obtain seed liquid;

[0114] (3) Production in a 5000 L fermentation tank: inoculating seed liquid into a fermentation tank filled with 3500 L of fermentation medium according to the inoculation amount of 8% volume ratio for culturing, wherein the fermentation medium comprises 40 g of bean cake powder, 50 g of bran, 60 g of corn flour, 10 g of dipotassium phosphate, 10 g of magnesium sulfate and 5 g of manganese sulfate according to the proportion for preparing to obtain the fermentation medium, adjusting the pH value to 7.8 by using sodium hydroxide solution, and fixing the volume to 1000 ml by using deionized water; controlling the rotation speed of the fermentation tank to be 800 rpm, controlling the aeration ratio to be 1.8 V/V.Math.min, controlling the pH to be 8 at the temperature of 30° C., culturing for 72 h, and obtaining fermentation liquid after the fermentation is finished;

[0115] (4) Carrying out ultrafiltration on the fermentation liquid obtained in step (3) by adopting a ceramic membrane with a pore diameter of 0.22  $\mu\text{m}$  to obtain *Bacillus subtilis* thalli and obtain enzyme liquid;

[0116] (5) An isomeric system: 5 ml of the enzyme solution was mixed with 114 g of fructose and 109 ml of deionized water, followed by reaction at 53° C. for 3.5 h. The psicose conversion was determined and calculated according to the method of example 2. The result shows that the conversion rate of the psicose produced by the technical scheme is 31.4%.

#### COMPARATIVE EXAMPLE 1

[0117] A method for producing psicose by fermentation and isomerization of *Bacillus subtilis* comprises the following specific implementation steps:

[0118] (1) Seed preparation: inoculating the *Bacillus subtilis* implanted with the *Closridium* gene prepared in example 1 into an LB medium, culturing for 24 h at the LB medium temperature of 37° C. and rotation speed of 250 rpm;

[0119] (2) Cultivation in a 50 L seed tank: inoculating the *Bacillus subtilis* obtained in step (1) into a seed tank filled with 35 L of LB culture medium according to the inoculation amount of 8% of the volume ratio for culturing, controlling the rotation speed of the seed tank to be 200 rpm, controlling the aeration ratio to be 0.3 V/V.Math.min, and culturing at the temperature of 30° C. for 24 h to obtain seed liquid;

[0120] (3) Production in a 5000 L fermentation tank: inoculating seed liquid into a fermentation tank filled with 3500 L of fermentation medium for culture according to the inoculation amount of 8% volume ratio, wherein the preparation method of the fermentation medium comprises the steps of mixing 40 g of bean cake powder, 50 g of bran, 60 g of corn flour, 10 g of dipotassium phosphate, 10 g of magnesium sulfate and 5 g of manganese sulfate, adjusting the pH to 7.8 by using a sodium hydroxide solution, and fixing the volume to 1000 ml by using deionized water; controlling the rotation speed of the fermentation tank to be 100 rpm, controlling the aeration ratio to be 1.8 V/V.Math.min, controlling the pH to be 8 at the temperature of 30° C., culturing for 72 h, and obtaining fermentation liquid after the fermentation is finished;

[0121] (4) Separating the *Bacillus subtilis* thalli from the fermentation liquid obtained in step (3) by adopting a ceramic membrane ultrafiltration method with a pore diameter of 0.22  $\mu\text{m}$ ;

[0122] (5) An isomeric system: 5 ml of *Bacillus subtilis* cells as an enzyme solution were mixed with 114 g of fructose and 109 ml of deionized water, followed by reaction at 53° C. for 3.5 h. The psicose conversion was determined and calculated according to the method of example 2. The result shows that the conversion rate of the psicose produced by the technical scheme is 18.4%.

#### COMPARATIVE EXAMPLE 2

[0123] A method for producing psicose by fermentation and isomerization of *Bacillus subtilis* comprises the following specific implementation steps:

[0124] (1) Seed preparation: inoculating the *Bacillus subtilis* implanted with the *Closridium* gene prepared in example 1 into an LB medium, culturing for 24 h at the LB medium temperature of 37° C. and rotation speed of 250 rpm;

[0125] (2) Cultivation in a 50 L seed tank: inoculating the *Bacillus subtilis* obtained in step (1) into a seed tank filled with 35 L of LB culture medium according to the inoculation amount of 8% of

the volume ratio for culturing, controlling the rotation speed of the seed tank to be 400 rpm, controlling the aeration ratio to be 0.45 V/V.Math.min, and culturing at the temperature of 30° C. for 24 h to obtain seed liquid;

[0126] (3) Production in a 5000 L fermentation tank: inoculating seed liquid into a fermentation tank filled with 3500 L of fermentation medium for culture according to the inoculation amount of 8% volume ratio, wherein the preparation method of the fermentation medium comprises the steps of mixing 40 g of bean cake powder, 50 g of bran, 60 g of corn flour, 10 g of dipotassium phosphate, 10 g of magnesium sulfate and 5 g of manganese sulfate, adjusting the pH to 7.8 by using a sodium hydroxide solution, and fixing the volume to 1000 ml by using deionized water; controlling the rotation speed of the fermentation tank to be 800 rpm, controlling the aeration ratio to be 3.4 V/V.Math.min, controlling the pH to be 8 at the temperature of 30° C., culturing for 72 h, and obtaining fermentation liquid after the fermentation is finished;

[0127] (4) Carrying out ultrafiltration on the fermentation liquid obtained in step (3) by adopting a ceramic membrane with a pore diameter of 0.22 nm to obtain *Bacillus subtilis* thalli and obtain enzyme liquid;

[0128] (5) An isomeric system: 5 ml of the enzyme solution was mixed with 114 g of fructose and 109 ml of deionized water, followed by reaction at 53° C. for 3.5 h. The psicose conversion was determined and calculated according to the method of example 2. The result shows that the conversion rate of the psicose produced by the technical scheme is 21.4%.

#### EXAMPLE 5

##### Enzyme Activity Assay

[0129] According to the conversion rate of the psicose in the examples 2-4 and the comparative examples 1-2, the enzyme activity of D-psicose-3-epimerase under different fermentation processes is calculated, and specifically calculated according to a formula I.

$$[00001] X = \frac{A \times 100 \times 9.5 \times 10^{-3} \times 10^6}{180.16 \times 10 \times 0.5} \quad \text{Formula I}$$

[0130] In formula 1: [0131] X is the enzyme activity of D-psicose-3-epimerase, and the unit is U/ml; [0132] A—peak area ratio of D-psicose in liquid chromatogram, unit is %; [0133] 100 g/L—fructose concentration, unit is g/L; [0134] 9.5 mL—fructose volume added to the reaction system, unit is milliliter (mL); [0135] 10.sup.-3—unit conversion factor, converting ml to L; [0136] 10.sup.6—unit conversion factor, converting mol to μmol; [0137] 180.16 cmolar mass of D-psicose, g/mol; [0138] 10—reaction time, unit is minute (min); [0139] 0.5—the amount of enzyme solution to be tested, unit is milliliter (mL).

[0140] The results are shown in Table 5.

TABLE-US-00005 TABLE 5 Enzyme Activity results for D-psicose-3-epimerase from different fermentation Processes Treat- Comparative Comparative ment Example 2 Example 3 Example 4 Example 1 Example 2 Enzyme 309.115 314.39 331.27 194.12 225.77 activity (U/mL)

[0141] As can be seen from the results of the above examples and comparative examples, through the exploration of fermentation process, this invention enables the high enzyme activity of D-psicose-3-epimerase to be obtained from recombinant *Bacillus subtilis* by a single feeding during fermentation, and the yield of psicose is ensured in fructose isomerization catalytic reaction. Compared with the experimental result of a comparative example, the ventilation volume, fermentation temperature, fermentation time, rotation speed and the culture medium are strictly controlled in the seed liquid culture and fermentation culture stages, so that higher enzyme activity is ensured, and the yield of the psicose is further improved.

[0142] The above description of the embodiments is only intended to facilitate the understanding of the method of the invention and its core idea. It should be noted that, for those skilled in the art, without departing from the principle of the present invention, it is possible to make various improvements and modifications to the present invention, and those improvements and modifications also fall within the scope of the claims of the present invention. Various

modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the invention. Thus, the present invention is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

## Claims

1. A method for producing psicose by fermentation and isomerization of *Bacillus subtilis* characterized by comprising the following steps: Inoculating recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing D-psicose 3-epimerase into a fermentation culture medium for fermentation culture to obtain a fermentation liquid; during the fermentation culture period, the aeration ratio is 0.1-2 V/V.Math.min; the rotation speed of the fermentation culture is 200-1000 rpm; Separating thalli from the fermentation liquid; Carrying out catalytic reaction on a substrate fructose by the thalli to obtain psicose.
  2. The method according to claim 1, wherein the fermentation medium comprises the following components: 10-40 g/L of bean cake powder, 20-50 g/L of bran, 30-60 g/L of corn flour, 1-10 g/L of dipotassium hydrogen phosphate, 1-10 g/L of magnesium sulfate and 0.1-5 g/L of manganese sulfate, wherein the pH value of the fermentation medium is 6-8.
  3. The method according to claim 2, wherein the fermentation medium comprises the following components: 20-35 g/L of bean cake powder, 35-45 g/L of bran, 45-55 g/L of corn flour, 4-8 g/L of dipotassium hydrogen phosphate, 5-8 g/L of magnesium sulfate and 1-3 g/L of manganese sulfate, wherein the pH value of the fermentation medium is 6.5-7.5.
  4. The method according to claim 1, wherein the temperature of the fermentation culture is 30-37° C., and the time of the fermentation culture is 48-72 h; During fermentation, controlling the pH value of the fermentation system to be 6-8; The inoculation amount of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is 1-10%.
  5. The method according to claim 1, wherein the system of the catalytic reaction is as follows: 1-5 ml of the thalli suspension, 50-150 g of fructose and 50-200 ml of water; The thalli concentration of the thalli suspension is  $(30-48) \times 10^9$  CFU/ml.
  6. The method according to claim 1, wherein the temperature of the catalytic reaction is 50-60° C.; The time of the catalytic reaction is 2-4 h.
  7. The method according to claim 1, wherein the culture medium of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing D-psicose 3-epimerase is LB medium; The culture temperature of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is 30-37° C.; The culture time of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is 20-24 h; The culture rotation speed of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is 200-400 rpm; When the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is cultured, the aeration ratio is 0.1-1 V/V.Math.min; The inoculation amount of the recombinant *Bacillus subtilis* 1A751-DPE seed liquid for expressing the D-psicose 3-epimerase is 1-10%.
  8. The method according to claim 1, wherein the method for separating the thalli is ceramic membrane ultrafiltration; The pore diameter of the ceramic membrane of the ceramic membrane ultrafiltration is 0.22  $\mu$ m or less.
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